

Tactical DNA: Coaches, Styles, and Player Roles from Football Passing Networks

Final Project Report for EECS 4414 Information Networks

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Abstract

Football tactics are debated qualitatively but rarely measured. We treat passing networks (directed weighted graphs whose edges count completed passes between players) as *tactical fingerprints* and use their structural signatures for three inference tasks: identifying a team’s coach from network features, discovering tactical archetypes by unsupervised clustering, and embedding players into a structural role space. Using StatsBomb open event data, we build per-match passing networks and a feature pipeline spanning centrality, global structure, and network motifs. Across FC Barcelona’s coaching eras, the 2018 and 2022 World Cups, and Bayern Munich, we report three findings. Coach identity is only modestly recoverable from passing structure, and under both leakage-free cross-validation and a cross-context transfer test the signal proves largely bound to the club and its roster rather than to the coach. Unsupervised clustering recovers a single dominant possession-versus-direct axis, with tactical styles forming a continuum rather than discrete archetypes. A structural role embedding recovers on-pitch position strongly, with 0.88 position purity among the five nearest neighbours (three times the random baseline), and surfaces tactically coherent playmaker analogues. Throughout, centrality rather than motif structure carries most of the discriminative signal.

Keywords

network analysis, football analytics, passing networks, social network analysis, motif analysis, sports analytics

1 Introduction and Motivation

Football tactics are extensively debated in informal media but rarely quantified with rigor. Coaches, pundits, and fans speak of “possession football,” “gegenpressing,” or a team’s “style of play” as if these were measurable properties, yet such labels are typically applied qualitatively. Passing networks (directed weighted graphs whose nodes represent players and whose edge weights count completed passes from one player to another) provide a principled way to encode tactical structure as a measurable mathematical object [4, 9]. A growing body of work has shown that passing-network properties relate to team performance and offensive efficiency [2, 6], but most studies are *descriptive*: they characterize a single team or compare a small number of matches, without using network signatures as features for higher-level inference tasks.

We close this gap by treating passing networks as tactical fingerprints and using them to answer three concrete inferential questions, defined formally in Section 2: can we identify a team’s coach from the structure of its passing alone (*coach identification*); do

matches cluster into recognizable tactical schools without supervision (*archetype discovery*); and can players be embedded into a structural role space that recovers their on-pitch position and surfaces structural “successors” to canonical playmakers (*player role embedding*)? The central motivating hypothesis is that a coach imposes a reproducible structural signature on a team’s passing, one strong enough to be recovered by a classifier and, potentially, to persist when the coach changes clubs. The problem is important both as a test of whether qualitative tactical language has a measurable structural basis and for its applications in opponent scouting, recruitment (finding players who fill a structural role), and the automated summarization of playing style.

2 Problem Definition

Notation. Let $G_m = (V_m, E_m, w_m)$ denote the passing network of a team in match m , where V_m is the set of starting players, $E_m \subseteq V_m \times V_m$ is the set of directed pass relations, and $w_m : E_m \rightarrow \mathbb{Z}^+$ assigns each edge the number of completed passes from one player to another. To keep graphs comparable we fix $|V_m| = 11$ (the starting eleven; see Section 4.1). Let $\phi(G_m) \in \mathbb{R}^d$ be a feature embedding of G_m derived from its structural properties (centrality distributions, global metrics, and motif counts), and for Problem 3 let $\psi(p) \in \mathbb{R}^{d'}$ be an aggregated ego-network signature of player p across all matches in which they appeared.

Problem 1 (Coach Identification). Given a labeled set $\{(G_m, c_m)\}$ where $c_m \in C$ is the coach of the team in match m , learn a classifier $f : \mathbb{R}^d \rightarrow C$ minimizing the expected misclassification rate, equivalently maximizing accuracy and per-class macro-F1, such that $f(\phi(G_m))$ predicts c_m significantly above a naive baseline. A key sub-question is whether a coach’s signature is *portable*: does it persist when the coach moves to a different team, or is it bound to the club and its roster? The marquee case is Pep Guardiola, from Barcelona (2008/09–2011/12) to Bayern Munich (2015/16); we extend the same test to coaches who subsequently led national teams (Section 5).

Problem 2 (Tactical Archetype Discovery). Given an unlabeled set $\{G_m\}$, partition the matches into clusters $\{C_1, \dots, C_k\}$ that optimize an internal cluster-quality objective (silhouette, modularity). We then evaluate, *a posteriori*, whether the discovered clusters correspond to known tactical schools such as possession-based play, gegenpressing, or low-block counter-attack.

Problem 3 (Player Role Embedding). For each player p , learn a low-dimensional embedding $\Phi : \mathbb{R}^{d'} \rightarrow \mathbb{R}^2$ of the ego-network signature $\psi(p)$ that preserves structural similarity, evaluated by *position purity*: the fraction of each player’s k nearest neighbors in

the embedding that share their on-pitch position. We complement this with qualitative case studies, identifying structural successors to canonical playmakers such as Xavi Hernández, Andrés Iniesta, and Luka Modrić.

2.1 Hardness

The three problems are hard along distinct axes. *Problem 1* is a statistical learning problem in a high-dimensional, small-sample, and class-imbalanced regime (coaches with single-season tenures contribute few matches), where the principal difficulty is *identifiability*: distinguishing a genuine coach signal from the confounds of team identity, squad personnel, and opponent. Its feature-extraction subroutines also carry computational cost: betweenness centrality is computed in $O(|V||E|)$ time by Brandes’ algorithm, and exact motif (subgraph) counting is intractable in general, being closely tied to subgraph isomorphism (NP-complete) and #P-hard for counting; we sidestep this by fixing the motif size to three nodes, for which the triadic census is polynomial. *Problem 2* is hard in the optimization sense: minimizing the within-cluster sum of squares for k -means is NP-hard [1], and maximizing modularity is likewise NP-hard [3], so we necessarily rely on heuristics (Lloyd’s algorithm, agglomerative merging) with multiple restarts rather than exact optimization. *Problem 3* requires optimizing a non-convex embedding objective; neighbor-preserving methods such as UMAP have no global-optimality guarantee, and a naive all-pairs structural-similarity computation is $O(n^2)$ in the number of players. We discuss the practical mitigations for each in Sections 4 and 5.

3 Related Work

Passing networks as a model of tactics. López Peña and Touchette [9] introduced a network-theoretic treatment of football strategy, using centrality and flow measures on passing graphs to characterize team play; Buldú et al. [4] extended this with a multilayer, space- and time-aware analysis of passing dynamics. Both establish that passing networks encode tactically meaningful structure. Our work builds on their structural descriptors but differs in aim: rather than describing a single team or match, we use the descriptors as features $\phi(G_m)$ for supervised and unsupervised *inference* across hundreds of matches.

From description to inference. The systematic review of Alves et al. [2] catalogues social-network analysis in football and shows the literature is dominated by descriptive performance and tactical applications. This positions our contribution precisely: we treat the same network signatures as inputs to higher-level inference tasks (identification, clustering, embedding) that the descriptive body largely leaves unexplored.

Motifs. Milo et al. [8] established network motifs as the recurring building blocks of complex networks. Li et al. [6] applied triadic-motif analysis to football passing networks across eighteen competitions, finding that motifs with more bidirectional (back-and-forth) links are the most significant and that teams tend to recirculate the ball rather than play direct forward chains. This motivates our inclusion of motif frequencies as a feature family, whose discriminative value for Problems 1–2 we test empirically.

High-performance and temporal structure. Chacoma [5] identifies and optimizes high-performance passing networks and

studies their temporal evolution within matches, motivating our planned temporal extension (per-segment networks).

Embedding methods. For the low-dimensional projections in Problems 2 and 3 we adopt UMAP [7], which preserves local neighborhood structure better than linear projections; we pair it with PCA for interpretability. All data are drawn from the StatsBomb open release [10].

4 Methodology

4.1 Data and Network Construction

We use the publicly available StatsBomb Open Data [10], which provides professional-grade event-level records, each annotated with a timestamp, pitch location on a standardized 120×80 grid, the players involved, and an outcome. For each match m we construct each team’s passing network G_m over the *eleven-starter window*: all events preceding the first substitution by *either* team. This is the conservative choice: it guarantees that all twenty-two starters are on the pitch throughout, at the cost of ending the window early when the opponent substitutes first (in 280 of 515 Barcelona matches with a substitution), trimming some otherwise-valid passing. Following established practice [4, 9], restricting to the eleven starters keeps $|V_m| = 11$ across all matches, removing the size confound that would otherwise distort centrality and motif statistics.

Starting elevens are read from the match lineups (players whose start reason is “Starting XI”) and cross-checked against the Starting XI tactics event.¹ An edge (u, v) is added for each completed open-play pass from player u to player v , weighted by the number of such passes; passes whose type is a throw-in, corner, free kick, goal kick, or kick-off are excluded so that edges reflect open-play structure, while passes immediately following a recovery or interception are retained as open play. We verified that the resulting corpus has no self-loops, all edge weights at least one, and exactly eleven nodes in every network.

4.2 Network Features

The feature embedding $\phi(G_m)$ aggregates three families of structural descriptors, each shown in prior work to carry tactical signal:

Centrality measures. For each node $v \in V_m$ we compute weighted in-degree and out-degree, betweenness centrality, eigenvector centrality, and PageRank. From the resulting per-node distributions we extract summary statistics (mean, variance, skewness, Gini coefficient), yielding a centralization profile that captures how hierarchically organized the team’s passing is [4].

Global structural metrics. Network density, weighted clustering coefficient, average shortest-path length, algebraic connectivity (the second-smallest eigenvalue of the graph Laplacian), and the largest eigenvalue of the adjacency matrix (spectral radius). Together these capture how cohesive a team’s passing organization is.

Network motifs. Motifs are small recurring subgraph patterns whose frequencies characterize the local structural “vocabulary” of a graph [8]. We enumerate the connected directed three-node motifs via the triadic census and record their normalized frequencies. As

¹One match (Granada vs. Barcelona, 2012/13) carries a lineups tagging error in which only two players are marked as starters; we recover the correct eleven from the intact Starting XI event.

discussed in Section 2, fixing the motif size to three nodes keeps this computation polynomial [6].

For Problem 3, we construct a separate per-player feature vector $\psi(p) \in \mathbb{R}^{d'}$ from each player’s ego-network properties aggregated across all matches in which they appeared: ego degree, ego density, ego clustering coefficient, weighted in/out-degree, motif-position frequencies (e.g., how often the player occupies the middle of a chain motif), and the spatial dispersion of pass endpoints.

Computational choices. Three definitions warrant note. For betweenness and weighted shortest-path computations we set the distance of an edge to the reciprocal of its pass count ($1/w$), since more passes denote a stronger and effectively shorter tie; using the raw weight as a distance would invert this relationship. Average shortest-path length is measured on the largest strongly connected component, as a handful of networks built from very short windows are not strongly connected. Eigenvector centrality is computed on the undirected weighted graph, whose symmetric adjacency matrix guarantees a real, well-defined leading eigenvector and which is the established convention in the passing-network literature [4, 9].

4.3 Algorithms

For **Problem 1**, we train three supervised classifiers on $\phi(G_m)$ with coach identity as the target: logistic regression as an interpretable linear baseline, a random forest to capture non-linear feature interactions, and XGBoost as a gradient-boosted ensemble. All are evaluated with stratified 5-fold cross-validation and class-weighted objectives to handle the imbalance arising from unequal match counts per coach.

For **Problem 2**, we apply k -means and hierarchical agglomerative clustering on $\phi(G_m)$, selecting k via the silhouette score and the elbow method, and visualize the result with UMAP [7] projections interpreted against known tactical schools.

For **Problem 3**, we project $\psi(p)$ into a two-dimensional role space using PCA (interpretable, linear) and UMAP (non-linear, neighborhood preserving), then use k -nearest-neighbor retrieval to identify structurally similar players.

All analysis is implemented in Python with NetworkX for graph construction and feature computation, scikit-learn for classical models and clustering, XGBoost for gradient boosting, and umap-learn for the projections. The full codebase and reproducible scripts are included with this submission.

4.4 Limitations and Difficulties

Three difficulties shape the approach. (i) Coach labels are not present in the event data and must be assembled from external records and aligned to match dates. (ii) Motif and centrality statistics are sensitive to graph size, which we neutralize by fixing $|V_m| = 11$. (iii) The richest within-club coaching variation in the open data lies with FC Barcelona; other clubs appear only as opponents (Section 5), so the multi-team experiments draw their tactical diversity from the World Cup national teams rather than from additional clubs. (iv) A small number of networks (16 of 517 with fewer than 100 passes, 5 with fewer than 50) arise from very early substitutions and the resulting short windows; we retain them but flag them as low-sample.

Table 1: Per-season composition of the corpus (FC Barcelona, La Liga 2005/06–2020/21). Mean passes and edges are per match over the eleven-starter window.

Season	Matches	Mean passes	Mean edges
2005/06	17	246.2	74.5
2006/07	26	273.9	74.5
2007/08	27	255.7	75.0
2008/09	31	256.8	73.4
2009/10	35	298.7	77.8
2010/11	33	351.9	80.4
2011/12	37	346.3	81.9
2012/13	32	344.3	77.7
2013/14	31	285.7	74.3
2014/15	38	366.6	83.8
2015/16	38	298.6	75.1
2016/17	34	300.9	77.7
2017/18	36	319.4	80.2
2018/19	34	323.1	79.8
2019/20	33	354.7	83.5
2020/21	35	338.1	81.7
Total	517		

5 Evaluation

5.1 Datasets

Our target corpus spans three competitive contexts: the La Liga “Big Three Golden Era,” Pep Guardiola’s 2015/16 Bundesliga season at Bayern Munich, and the men’s FIFA World Cups of 2018 and 2022. The StatsBomb open release covers all three contexts, though unevenly: it provides every FC Barcelona La Liga match and both recent men’s World Cups in full, but its 2015/16 Bundesliga release is centred on Bayer Leverkusen, so Guardiola’s Bayern Munich appears only as Leverkusen’s opponent (just two matches), which sharply limits the transfer experiment below. We anchor the corpus on the portion with the richest within-club coaching variation, **FC Barcelona across its successive coaching eras**. We retain the sixteen La Liga seasons from 2005/06 to 2020/21 with adequate coverage (two earlier seasons with fewer than ten Barcelona matches were excluded as too thin for reliable per-season statistics), yielding **517 team-match passing networks**. Restricting to a single club isolates the coach signal from the confound of team identity, directly addressing the central question of Problem 1. Table 1 reports the per-season composition of the corpus. For Problems 2 and 3 we add the World Cup national teams, which inject diverse tactical contexts; Real Madrid, Atlético Madrid, and Bayern Munich appear in the open data only as opponents, so the cross-context transfer test below necessarily rests on just two Guardiola-Bayern matches. In total the corpus comprises **775 team-match networks across 42 teams** (517 Barcelona, 256 World Cup, 2 Bayern): Problem 1 uses the Barcelona subset, while Problems 2 and 3 use the full corpus.

Table 2: Summary statistics of the FC Barcelona passing-network corpus (mean $\pm$ std across matches).

Metric	Mean $\pm$ Std
Number of edges	78.52 $\pm$ 11.17
Density	0.714 $\pm$ 0.102
Total completed passes	314.23 $\pm$ 110.58
Weighted clustering coefficient	0.181 $\pm$ 0.040
Avg. shortest-path length ($d = 1/w$)	0.400 $\pm$ 0.215
Avg. shortest-path length (hops)	1.296 $\pm$ 0.134
Algebraic connectivity (Fiedler)	16.21 $\pm$ 7.89
Spectral radius	32.92 $\pm$ 11.83

5.2 Summary Statistics of the Passing-Network Corpus

Across the 517 networks, Barcelona’s passing is dense and triangle-rich (mean density 0.71 ± 0.10 , mean weighted clustering 0.18 ± 0.04), consistent with the short-passing, possession-oriented style the side is known for. Per-match volume varies widely, from 18 to 660 completed open-play passes (median 309), because the window length depends on when the first substitution falls. Table 2 reports the global metrics over the full Barcelona corpus.

The per-season breakdown reveals a measurable tactical signature. Cohesion metrics are lowest in the mid-2000s rebuilding seasons (density around 0.67 – 0.68 and algebraic connectivity around 11 – 13 in 2005/06–2008/09) and rise markedly through the trophy-laden 2010s. Individual metrics peak in standout seasons: weighted clustering at 0.207 in Guardiola’s 2011/12, density at 0.761 in Luis Enrique’s 2014/15 treble side, and algebraic connectivity at 23.5 in 2019/20. That the networks become quantifiably more cohesive in the team’s strongest periods is an early signal supporting the project’s central hypothesis that tactical quality leaves a measurable structural footprint, though we note this trend co-varies with passing volume (Table 1) and so partly reflects the greater possession of dominant sides rather than tactical quality alone. Figure 1 shows a representative network: Barcelona’s 5–0 win over Real Madrid in November 2010, with Xavi the highest-PageRank node (the passing metronome), exactly the hierarchy the qualitative account would predict.

5.3 Evaluation Protocol

Problem 1. We report accuracy and per-class macro-F1 under stratified 5-fold cross-validation and publish a confusion matrix exposing which coaches are confused with which. The principal baseline is a majority-class classifier; the stronger check that the signal exceeds team identity is the cross-context transfer test, which asks whether a coach is still recognized when managing a different squad. That experiment trains on Barcelona and tests three of its coaches with their non-Barcelona teams (Guardiola at Bayern, Luis Enrique with Spain, Martino with Mexico). A further analysis ablates feature families (centrality, motifs, combined) to identify the most discriminative descriptors.

Problem 2. We report silhouette, modularity, and average intra- vs. inter-cluster distance across candidate k , then manually inspect

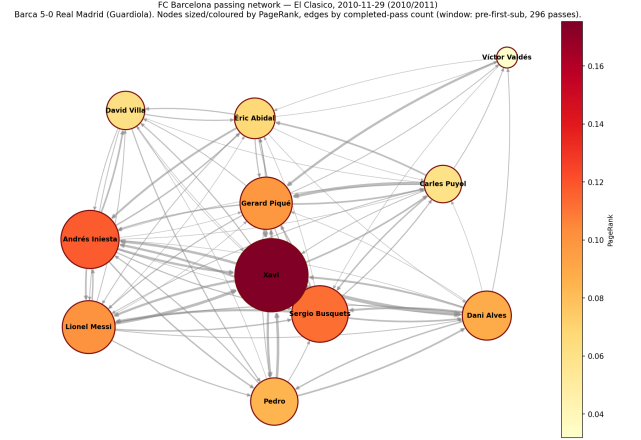


Figure 1: FC Barcelona’s passing network in the 2010-11-29 El Clásico (5–0, Guardiola). Nodes are the eleven starters, sized and coloured by PageRank; directed edges are weighted by completed open-play pass count during the pre-substitution window. Xavi (centre) is the top-PageRank node, consistent with his role as the side’s passing metronome.

each cluster against human-assigned tactical categories (possession, gegenpressing, low-block counter, direct). A UMAP projection annotated with landmark matches (marquee Clásicos from the 2010–2013 Pep-vs-Mourinho era and the 2022 World Cup Final) serves as the principal qualitative figure.

Problem 3. We test whether nearest-neighbor retrieval respects position (a midfielder’s neighbors should be predominantly midfielders), reporting position purity at $k = 5$ and $k = 10$ against a random-placement baseline, complemented by playmaker-successor case studies (Xavi, Iniesta, Busquets, Modrić).

5.4 Results: Coach Identification

Problem 1 is an eight-class coach-identification task over the 517 FC Barcelona networks, with per-match labels assigned by date.² The class distribution is heavily skewed (Table 4): Guardiola contributes 136 matches while three coaches held the post for only a single season.

Against a majority-class baseline of 0.263 , a class-weighted Random Forest reaches 0.358 ± 0.031 accuracy (macro-F1 0.229) under stratified 5-fold cross-validation, a $+0.095$ improvement; XGBoost follows at 0.298 and logistic regression trails at 0.244 (Table 5). An ablation (Table 6) shows that centrality features carry most of the discriminative signal for every model (random forest 0.321 alone, close to its 0.358 combined), while motif frequencies are weak in isolation (random forest 0.242 , barely above baseline). The confusion matrix (Figure 2) shows that Guardiola’s signature is by far the most distinctive (0.71 recall, 0.51 F1), whereas the three single-season coaches (Vilanova, Martino, Koeman) are essentially unidentifiable and collapse into the majority class.

²The 2019/20 season is split at Ernesto Valverde’s dismissal (13 January 2020) into his and Quique Setién’s spells, a more accurate ground truth than a season-level mapping. Xavi’s 2021/22–2023/24 tenure falls outside the free data, which ends at 2020/21.

Table 3: Cross-context transfer: a Barcelona-trained classifier predicting the same coach managing a different team. Cells are correct-recovery counts; the mean is over the three models.

Coach	Tested on	n	LR	RF	XGB	Mean
Guardiola	Bayern 2015/16	2	0/2	0/2	0/2	0.00
Luis Enrique	Spain, WC 2022	4	1/4	1/4	3/4	0.42
Martino	Mexico, WC 2022	3	1/3	0/3	0/3	0.11

Crucially, replacing random folds with a leakage-free *season-grouped* cross-validation (in which every season is kept intact within a single fold) drops Random Forest accuracy to 0.250 and macro-F1 to 0.113. Because each coach owns contiguous seasons with year-to-year roster turnover, random folds let same-season matches leak across train and test, so the model partly learns *which season* a match belongs to rather than *which coach*; the 0.108 accuracy gap quantifies this inflation. Under this leakage-free protocol all three models hover near the 0.263 baseline, with XGBoost generalizing marginally best (0.256 accuracy, 0.120 macro-F1). The honest reading is that network structure carries a measurable but modest coach signal above baseline, that much of the naive-CV gain is season-specific memorization, and that identifying a coach from a single season of matches is inherently hard.

Cross-context transfer. The marquee question is whether a coach’s signature follows the coach to a new team. We trained on the full Barcelona corpus and tested three of its coaches in a different context: Pep Guardiola at Bayern Munich (2015/16), Luis Enrique with Spain, and Gerardo Martino with Mexico, the latter two at the 2022 World Cup (Table 3). The open Bundesliga release is centred on Bayer Leverkusen, so Bayern appears only as Leverkusen’s opponent (just two Guardiola networks); the World Cup supplies seven further national-team networks for the other two coaches. Averaged across the three coaches, mean recovery is about 0.18, only marginally above the 0.125 uniform baseline, so the signature is on the whole *not* portable. Guardiola is the clearest case, recovered 0/2, his Bayern networks reading as other Barcelona coaches, exactly as the season-grouped result predicts. The one exception is Luis Enrique, whose possession identity partially survives to Spain (mean 0.42, with XGBoost recovering three of four matches), hinting that a sufficiently distinctive style may leave a more portable trace. The samples are small ($n \leq 4$ per coach) and the Enrique effect is model-dependent, so we read the result as suggestive: the dominant signal is club- and roster-bound, with stylistic extremity a plausible moderator worth testing on a larger corpus.

5.5 Results: Tactical Archetypes

Turning to Problem 2, we cluster the merged corpus (774 networks, after dropping one degenerate two-pass network) on the standardized feature embedding, comparing k -means and Ward agglomerative clustering for $k = 2, \dots, 10$. The silhouette score peaks sharply at $k = 2$ for both methods (0.245 for k -means, 0.197 for Ward) and the inertia elbow is smooth with no pronounced knee, so we

Table 4: Coach label distribution (FC Barcelona, La Liga 2005/06–2020/21).

Coach	Matches	Seasons
Guardiola	136	2008/09–2011/12
Luis Enrique	110	2014/15–2016/17
Valverde	84	2017/18–2019/20 (pre-Jan)
Rijkaard	70	2005/06–2007/08
Koeman	35	2020/21
Vilanova	32	2012/13
Martino	31	2013/14
Setién	19	2019/20 (post-Jan)
Total	517	8 coaches

Table 5: Coach identification on FC Barcelona, under standard (random) and leakage-free (season-grouped) 5-fold cross-validation.

Model	CV	Accuracy	Macro-F1
Majority baseline	n/a	0.263	0.052
Logistic Reg.	random 5-fold	0.244 ± 0.012	0.215 ± 0.010
Random Forest	random 5-fold	0.358 ± 0.031	0.229 ± 0.048
XGBoost	random 5-fold	0.298 ± 0.014	0.225 ± 0.010
Logistic Reg.	season-grouped	0.187 ± 0.057	0.104 ± 0.045
Random Forest	season-grouped	0.250 ± 0.090	0.113 ± 0.038
XGBoost	season-grouped	0.256 ± 0.069	0.120 ± 0.044

Table 6: Feature-family ablation (accuracy, stratified 5-fold CV).

Feature set	LR	RF	XGBoost
Centrality-only	0.205	0.321	0.296
Motifs-only	0.172	0.242	0.230
Combined	0.244	0.358	0.298

take $k = 2$ as the natural partition (modularity 0.312, mean inter-/intra-cluster distance ratio 1.19). The two clusters separate along a single dominant axis, passing *density*. Cluster 1 (547 networks) is the high-density, triangle-rich possession archetype: it holds 435 of the 517 Barcelona networks together with possession-minded national teams (Spain, Portugal, Croatia). Cluster 0 (227 networks) is the lower-density, more direct archetype, dominated by World Cup sides such as Poland, Uruguay, France, and Morocco, alongside the 82 Barcelona matches (largely the mid-2000s Rijkaard rebuild) whose networks were less possession-heavy.

Two points stand out. First, the axis the clustering recovers without supervision is exactly the possession-versus-direct contrast the proposal hypothesized, and it cuts *across* competitions rather than merely separating leagues: possession-style national teams sit with Barcelona, while Barcelona’s least dominant era drifts toward the direct cluster. Second, the modest silhouette and knee-free elbow indicate that tactical styles form a *continuum* rather than cleanly separated archetypes; the UMAP projection (Figure 3)

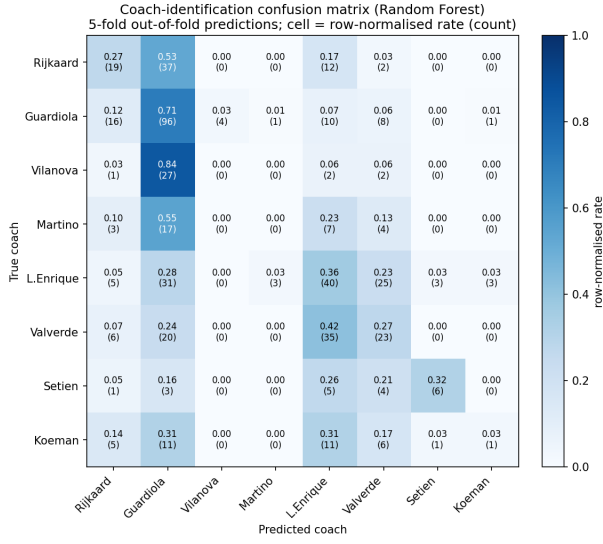


Figure 2: Row-normalized confusion matrix for coach identification (Random Forest, out-of-fold predictions). Guardiola is the most reliably recovered; single-season coaches collapse into the majority class.

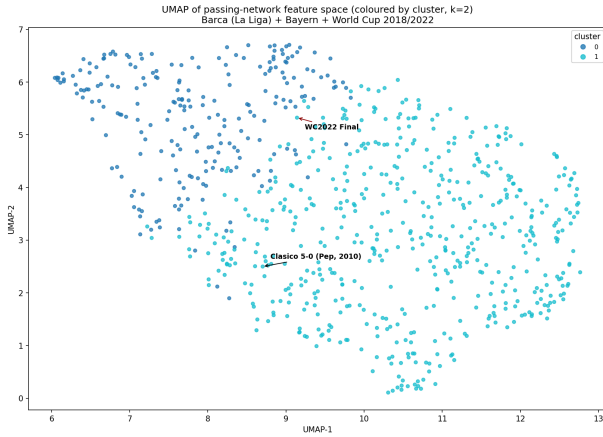


Figure 3: UMAP of the passing-network feature space, coloured by cluster ($k = 2$). The possession-heavy cluster (Barcelona with possession-minded national teams) and the more direct cluster occupy distinct regions joined by a gradient, consistent with tactical style being a continuum. Landmarks: the 2010 5–0 Clásico and the 2022 World Cup Final.

shows the two clusters in distinct regions joined by a gradient, with Guardiola’s 2010 5–0 Clásico deep in the possession region and the 2022 World Cup Final near the boundary. The chief caveat is compositional: Barcelona is 67% of the corpus, so the possession cluster is Barcelona-dominated by construction, and de-biasing would need other clubs as their own teams, which the open data does not provide.

Table 7: Position purity of the player role embedding (fraction of the k nearest neighbours sharing the player’s positional line), against a random baseline of 0.295.

Embedding space	$k = 5$	$k = 10$
Raw $\psi(p)$	0.883	0.856
UMAP (2D)	0.865	0.868
PCA (2D)	0.680	0.668
Random baseline	0.295	0.295

5.6 Results: Player Roles

Problem 3 embeds players rather than matches. For every player with at least five appearances (271 in total: 105 defenders, 79 midfielders, 61 forwards, 26 goalkeepers) we aggregate a 14-dimensional ego-network and spatial signature $\psi(p)$ and project it to two dimensions with PCA and UMAP, coloured by positional line. The embedding recovers on-pitch position strongly. Table 7 reports position purity, the fraction of each player’s k nearest neighbours sharing its line, against a random-placement baseline of 0.295: in the raw $\psi(p)$ space 0.88 of a player’s five nearest neighbours share its line, three times chance, and UMAP preserves this almost exactly in two dimensions (0.86), while PCA retains less (0.68) but still more than doubles the baseline. The UMAP map (Figure 4) shows goalkeepers as a fully separate island and forwards, midfielders, and defenders in distinct regions, with the deep playmakers clustered tightly at the midfield boundary.

The neighbour queries are tactically coherent. Xavi’s ten nearest players are all central midfielders (Thiago, Cesc Fàbregas, Pjanić, Arthur, Brozović, Frenkie de Jong, Busquets, Iniesta), the deep-lying creators one would name by hand. Busquets, a single pivot, correctly pulls in ball-playing centre-backs (Mascherano, Otamendi, Akanji) alongside other holding midfielders, capturing his habit of dropping between the defenders. Iniesta’s neighbours are his ball-circulating heirs at the club (Arthur, Kovačić, Cesc, Frenkie de Jong). Most relevant to the project’s recruitment motivation, Luka Modrić, who appears only for Croatia, is matched to box-to-box creators drawn from several different national teams and clubs (Bernardo Silva, Rakitić, Rodrigo De Paul), a genuine cross-context structural analogue. The one gap is Toni Kroos, absent because the open data lacks Real Madrid, which limits the Modrić/Kroos successor comparison to national-team data.

5.7 Limitations

Several limitations temper these results. The corpus is heavily skewed toward FC Barcelona (67% of networks), which inflates the possession cluster and narrows the transfer baseline; class imbalance in the coach labels is likewise severe, mitigated only partially by class-weighted objectives and stratified sampling. Motif and centrality statistics remain sensitive to graph size, controlled by fixing $|V_m| = 11$. UMAP preserves local structure better than PCA but is harder to interpret, so we report both throughout. Finally, a natural extension we leave to future work is a temporal one: per-segment networks within each match (for example 15-minute windows) that would capture within-game tactical adaptation [5].

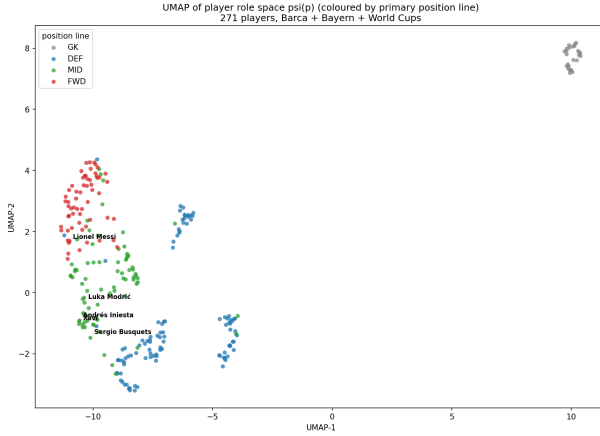


Figure 4: UMAP of the player role space $\psi(p)$ (271 players, coloured by positional line). Goalkeepers form a separate island; forwards, midfielders, and defenders occupy distinct regions; the deep playmakers (Xavi, Iniesta, Busquets, Modrić) cluster at the midfield boundary.

6 Conclusions and Future Work

This report asked whether the qualitative language of football tactics has a measurable structural basis, posing three inference tasks on passing networks built from StatsBomb open data over a feature embedding spanning centrality, global structure, and motifs. For Problem 1, coach identity is only modestly recoverable: a random forest reaches 0.358 against a 0.263 baseline under standard cross-validation, carried chiefly by centrality, but a leakage-free season-grouped evaluation drops to 0.250, showing that much of the naive gain is season-specific memorization and that single-season coaches are essentially unidentifiable. A cross-context transfer test over three coaches (Guardiola, Luis Enrique, and Martino with their non-Barcelona teams) points the same way: mean recovery across the three coaches is about 0.18, near the 0.125 chance level, so the signal is largely club- and roster-bound, the lone partial exception being Luis Enrique’s possession identity surviving to Spain. For Problem 2, unsupervised clustering independently recovers a possession-versus-direct axis as the primary tactical dimension, with the modest cluster separation indicating that styles vary along a continuum rather than forming discrete archetypes.

Problem 3 gives the clearest result: a structural role embedding recovers on-pitch position with 0.88 purity among the five nearest neighbours (three times chance), and its nearest-neighbour queries return tactically coherent playmaker analogues, including a genuine cross-context match for Luka Modrić. Two findings recur across the three tasks: centrality, not motif structure, carries the discriminative signal, and what looks like a coach “fingerprint” is substantially a property of the squad. The main limitation is the Barcelona-heavy corpus (67% of networks); because the open data contains Real Madrid and Atlético only as opponents, adding them as their own teams is the clearest path to de-biasing both the clustering and the transfer baseline. A per-segment temporal analysis of within-match adaptation remains the natural next step.

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